

Homework 9

Shanshan Cao

9.4.2

$$(1) \quad \psi = \frac{1}{\sqrt{a_0^3 \pi}} e^{-r/a_0}$$

$$\begin{aligned} \langle \psi | \psi \rangle &= \int_0^{2\pi} d\phi \int_0^\pi d(\cos\theta) \int_0^\infty dr \cdot r^2 \cdot \psi^* \psi \\ &= 4\pi \int_0^\infty dr \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot r^2 \\ &= \frac{4}{a_0^3} \int_0^\infty r^2 e^{-2r/a_0} dr \\ &= \frac{4}{a_0^3} \cdot \left(-\frac{a_0}{2}\right) \left(r^2 e^{-2r/a_0} \Big|_0^\infty - \int_0^\infty 2r e^{-2r/a_0} dr\right) \\ &= \frac{4}{a_0^3} \cdot \left(-\frac{a_0}{2}\right)^2 \cdot (-2) \cdot \left(r e^{-2r/a_0} \Big|_0^\infty - \int_0^\infty e^{-2r/a_0} dr\right) \\ &= \frac{4}{a_0^3} \cdot \left(-\frac{a_0}{2}\right)^3 \cdot 2 \cdot e^{-2r/a_0} \Big|_0^\infty \\ &= \frac{4}{a_0^3} \cdot \frac{a_0^3}{8} \cdot 2 \cdot 1 \\ &= 1. \end{aligned}$$

$$(2) \quad (\Delta X)^2 = \langle X^2 \rangle - \langle X \rangle^2$$

$$\langle X \rangle = \int r \sin\theta \cos\phi \cdot \psi^* \psi \cdot r^2 \sin\theta dr d\theta d\phi$$

$$\text{The } \phi \text{ part: } \int_0^{2\pi} d\phi \cos\phi d\phi = 0.$$

$$\text{Therefore, } \langle X \rangle = 0.$$

$$\begin{aligned} \langle X^2 \rangle &= \int r^2 \sin^2\theta \cos^2\phi \cdot \psi^* \psi \cdot r^2 \sin\theta dr d\theta d\phi \\ &= \left(\int_0^\infty \frac{1}{\pi a_0^3} \cdot r^4 e^{-2r/a_0} \cdot dr \right) \left(\int_0^\pi \sin^3\theta d\theta \right) \left(\int_0^{2\pi} \cos^2\phi d\phi \right) \\ &= \frac{3}{4\pi} a_0^2 \cdot \frac{4}{3} \cdot \pi \\ &= a_0^2 \end{aligned}$$

$$\text{Thus, } (\Delta X)^2 = a_0^2 = \left(-\frac{\hbar^2}{me^2}\right)^2 \Rightarrow \Delta X = \frac{\hbar^2}{me^2}.$$

$$r^2 = X^2 + Y^2 + Z^2$$

By symmetry

$$\Rightarrow \langle r^2 \rangle = \langle X^2 \rangle + \langle Y^2 \rangle + \langle Z^2 \rangle \stackrel{\downarrow}{=} 3 \langle X^2 \rangle = 3 (\Delta X)^2$$

$$\Rightarrow (\Delta X)^2 = \frac{1}{3} \langle r^2 \rangle.$$

It might be easier to compute $\langle r^2 \rangle$ first.

$$\begin{aligned}
 \langle r^2 \rangle &= 4\pi \int_0^\infty r^2 \cdot \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot r^2 dr \\
 &= 4\pi \cdot \frac{1}{\pi a_0^3} \int_0^\infty \left(\frac{2r}{a_0}\right)^4 e^{-2r/a_0} d\left(\frac{2r}{a_0}\right) \cdot \left(\frac{a_0}{2}\right)^5 \\
 &= 4\pi \cdot \frac{1}{\pi a_0^3} \cdot \left(\frac{a_0}{2}\right)^5 \cdot \Gamma(5) \\
 &= 4 \cdot a_0^2 \cdot \frac{1}{32} \cdot 4! \\
 &= 3a_0^2
 \end{aligned}$$

$$\Rightarrow (\Delta X)^2 = \frac{1}{3} \langle r^2 \rangle = a_0^2$$

$$\begin{aligned}
 (3) \quad \left\langle \frac{1}{r} \right\rangle &= 4\pi \int_0^\infty \frac{1}{r} \cdot \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot r^2 dr \\
 &= 4\pi \cdot \frac{1}{\pi a_0^3} \int_0^\infty \frac{2r}{a_0} e^{-2r/a_0} d\left(\frac{2r}{a_0}\right) \cdot \left(\frac{a_0}{2}\right)^2 \\
 &= 4\pi \cdot \frac{1}{\pi a_0^3} \cdot \frac{a_0^2}{4} \cdot 1! \\
 &= \frac{1}{a_0}
 \end{aligned}$$

$$\begin{aligned}
 \langle r \rangle &= 4\pi \int_0^\infty r \cdot \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot r^2 dr \\
 &= 4\pi \cdot \frac{1}{\pi a_0^3} \int_0^\infty \left(\frac{2r}{a_0}\right)^3 e^{-2r/a_0} d\left(\frac{2r}{a_0}\right) \cdot \left(\frac{a_0}{2}\right)^4 \\
 &= 4\pi \cdot \frac{1}{\pi a_0^3} \cdot \frac{a_0^4}{16} \cdot 3! \\
 &= \frac{3}{2} a_0
 \end{aligned}$$

$$\Rightarrow \frac{1}{\langle r \rangle} = \frac{2}{3a_0}$$

$$\text{Therefore, } \left\langle \frac{1}{r} \right\rangle \approx \frac{1}{\langle r \rangle} \approx \frac{1}{a_0} = \frac{me^2}{\hbar^2}.$$

10.1.1.

$$\begin{aligned}
 (1) \quad & [\Omega_1^{(1)} \otimes I^{(2)}, I^{(1)} \otimes \Lambda_2^{(2)}] |x_1\rangle \otimes |x_2\rangle \\
 &= (\Omega_1^{(1)} \otimes I^{(2)})(I^{(1)} \otimes \Lambda_2^{(2)})(|x_1\rangle \otimes |x_2\rangle) - (I^{(1)} \otimes \Lambda_2^{(2)})(\Omega_1^{(1)} \otimes I^{(2)})(|x_1\rangle \otimes |x_2\rangle) \\
 &= (\Omega_1^{(1)} \otimes I^{(2)})(|x_1\rangle \otimes |\Lambda_2^{(2)} x_2\rangle) - (I^{(1)} \otimes \Lambda_2^{(2)})(|\Omega_1^{(1)} x_1\rangle \otimes |x_2\rangle) \\
 &= |\Omega_1^{(1)} x_1\rangle \otimes |\Lambda_2^{(2)} x_2\rangle - |\Omega_1^{(1)} x_1\rangle \otimes |\Lambda_2^{(2)} x_2\rangle \\
 &= 0
 \end{aligned}$$

$$\Rightarrow [\Omega_1^{(1)} \otimes I^{(2)}, I^{(1)} \otimes \Lambda_2^{(2)}] = 0.$$

$$\begin{aligned}
(2) \quad & (\Omega_1^{(1)} \otimes T_2^{(2)})(\theta_1^{(1)} \otimes \Lambda_2^{(2)})(|\chi_1\rangle \otimes |\chi_2\rangle) \\
&= (\Omega_1^{(1)} \otimes T_2^{(2)})(|\theta_1^{(1)}\chi_1\rangle \otimes |\Lambda_2^{(2)}\chi_2\rangle) \\
&= |\Omega_1^{(1)}\theta_1^{(1)}\chi_1\rangle \otimes |T_2^{(2)}\Lambda_2^{(2)}\chi_2\rangle \\
&= (\Omega_1^{(1)}\theta_1^{(1)}) \otimes (T_2^{(2)}\Lambda_2^{(2)})(|\chi_1\rangle \otimes |\chi_2\rangle) \\
&= (\Omega\theta)_1^{(1)} \otimes (T\Lambda)_2^{(2)}(|\chi_1\rangle \otimes |\chi_2\rangle) \\
&\Rightarrow (\Omega_1^{(1)} \otimes T_2^{(2)})(\theta_1^{(1)} \otimes \Lambda_2^{(2)}) = (\Omega\theta)_1^{(1)} \otimes (T\Lambda)_2^{(2)}.
\end{aligned}$$

$$\begin{aligned}
(3) \quad & [\Omega_1^{(1) \otimes (2)}, \Lambda_1^{(1) \otimes (2)}] \\
&= [\Omega_1^{(1)} \otimes I^{(2)}, \Lambda_1^{(1)} \otimes I^{(2)}] \\
&= (\Omega_1^{(1)} \otimes I^{(2)})(\Lambda_1^{(1)} \otimes I^{(2)}) - (\Lambda_1^{(1)} \otimes I^{(2)})(\Omega_1^{(1)} \otimes I^{(2)}) \\
(\text{by part (2)}) \quad &= (\Omega_1^{(1)}\Lambda_1^{(1)}) \otimes I^{(2)} - (\Lambda_1^{(1)}\Omega_1^{(1)}) \otimes I^{(2)} \\
(\text{by linearity}) \quad &= (\Omega_1^{(1)}\Lambda_1^{(1)} - \Lambda_1^{(1)}\Omega_1^{(1)}) \otimes I^{(2)} \\
&= [\Omega_1^{(1)}, \Lambda_1^{(1)}] \otimes I^{(2)} \\
&= T_1^{(1)} \otimes I^{(2)}.
\end{aligned}$$

Similarly, we can prove $[\Omega_2^{(1) \otimes (2)}, \Lambda_2^{(1) \otimes (2)}] = I^{(1)} \otimes T_2^{(2)}$ if $[\Omega_2^{(1)}, \Lambda_2^{(1)}] = T_2^{(1)}$.

$$\begin{aligned}
(4) \quad & (\Omega_1^{(1) \otimes (2)} + \Omega_2^{(1) \otimes (2)})^2 \\
&= (\Omega_1^{(1)} \otimes I^{(2)} + I^{(1)} \otimes \Omega_2^{(2)})^2 \\
&= (\Omega_1^{(1)} \otimes I^{(2)})(\Omega_1^{(1)} \otimes I^{(2)}) + (\Omega_1^{(1)} \otimes I^{(2)})(I^{(1)} \otimes \Omega_2^{(2)}) + (I^{(1)} \otimes \Omega_2^{(2)})(\Omega_1^{(1)} \otimes I^{(2)}) \\
&\quad + (I^{(1)} \otimes \Omega_2^{(2)})(I^{(1)} \otimes \Omega_2^{(2)}) \\
(\text{by part (2)}) \quad &= (\Omega_1^{(2)})^{(1)} \otimes I^{(2)} + \Omega_1^{(1)} \otimes \Omega_2^{(2)} + \Omega_1^{(1)} \otimes \Omega_2^{(2)} + I^{(1)} \otimes (\Omega_2^{(2)})^2 \\
&= (\Omega_1^{(2)})^{(1)} \otimes I^{(2)} + I^{(1)} \otimes (\Omega_2^{(2)})^2 + 2\Omega_1^{(1)} \otimes \Omega_2^{(2)}.
\end{aligned}$$

10.1.2.

* If $A_1^{(1)} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $B_2^{(2)} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, we can prove that

$$A_1^{(1)} \otimes B_2^{(2)} = \begin{matrix} & \begin{matrix} 11 & 12 & 21 & 22 \end{matrix} \\ \begin{matrix} 11 \\ 12 \\ 21 \\ 22 \end{matrix} & \begin{pmatrix} x & x & x & x \\ x & x & x & x \\ x & x & x & x \\ x & x & x & x \end{pmatrix} \end{matrix} = \begin{pmatrix} a_{11} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{12} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ a_{21} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{22} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \end{pmatrix}$$

Proof: The new matrix element

$$\begin{aligned}
 M_{i_1 i_2, j_1 j_2} &= (\langle i_1 | \otimes \langle i_2 |) (A_1^{(1)} \otimes B_2^{(2)}) (|j_1\rangle \otimes |j_2\rangle) \\
 &= (\langle i_1 | \otimes \langle i_2 |) (A_1^{(1)} |j_1\rangle \otimes B_2^{(2)} |j_2\rangle) \\
 &= \langle i_1 | A_1^{(1)} |j_1\rangle \langle i_2 | B_2^{(2)} |j_2\rangle \\
 &= A_1^{(1)}{}_{i_1 j_1} \cdot B_2^{(2)}{}_{i_2 j_2} \quad i_1, i_2, j_1, j_2 = 1 \text{ or } 2.
 \end{aligned}$$

This is equivalent to the matrix form above.

(1)

$$\sigma_1^{(1)} \otimes I^{(2)} = \sigma_1^{(1)} \otimes I^{(2)} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & b \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ c \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & d \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{pmatrix}$$

* We set $1 \leftrightarrow "+"$, $2 \leftrightarrow "-"$,
and use the statement above.

(2)

$$\sigma_2^{(1)} \otimes I^{(2)} = I^{(1)} \otimes \sigma_2^{(2)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

$$= \begin{pmatrix} 1 \begin{pmatrix} e & f \\ g & h \end{pmatrix} & 0 \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ 0 \begin{pmatrix} e & f \\ g & h \end{pmatrix} & 1 \begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{pmatrix}$$

$$(3) (\sigma_1 \sigma_2)^{(1)} \otimes I^{(2)} = \sigma_1^{(1)} \otimes \sigma_2^{(2)} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

$$= \begin{pmatrix} a \begin{pmatrix} e & f \\ g & h \end{pmatrix} & b \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ c \begin{pmatrix} e & f \\ g & h \end{pmatrix} & d \begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}$$

10.2.1.

Inside the box, $V = 0$. The Schrödinger Equation:

$$E \psi(x, y, z) = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(x, y, z)$$

Separation of variables:

$$\psi(x, y, z) = X(x)Y(y)Z(z)$$

$$\Rightarrow E X(x)Y(y)Z(z) = -\frac{\hbar^2}{2m} \cdot Y(y)Z(z) \frac{\partial^2}{\partial x^2} X(x) - \frac{\hbar^2}{2m} X(x)Z(z) \frac{\partial^2}{\partial y^2} Y(y) - \frac{\hbar^2}{2m} X(x)Y(y) \frac{\partial^2}{\partial z^2} Z(z)$$

Divide both sides by $X(x)Y(y)Z(z)$

$$\Rightarrow E = -\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2}{\partial x^2} X - \frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2}{\partial y^2} Y - \frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2}{\partial z^2} Z$$

The only way to satisfy this equation is to set

$$\begin{cases} -\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2}{\partial x^2} X = E_x \\ -\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2}{\partial y^2} Y = E_y \\ -\frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2}{\partial z^2} Z = E_z \end{cases} \quad E = E_x + E_y + E_z$$

We have got 3 one-dimensional Schrödinger Equation. The solutions are:

$$\begin{cases} X(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right), & E_x = \frac{\hbar^2 \pi^2}{2mL^2} n_x^2 \\ Y(y) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_y \pi y}{L}\right), & E_y = \frac{\hbar^2 \pi^2}{2mL^2} n_y^2 \\ Z(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_z \pi z}{L}\right), & E_z = \frac{\hbar^2 \pi^2}{2mL^2} n_z^2 \end{cases}$$

$$n_x, n_y, n_z = 1, 2, 3, \dots$$

Therefore,

$$\psi_E(x, y, z) = X_E(x)Y_E(y)Z_E(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right) \sqrt{\frac{2}{L}} \sin\left(\frac{n_y \pi y}{L}\right) \sqrt{\frac{2}{L}} \sin\left(\frac{n_z \pi z}{L}\right),$$

$$E = E_x + E_y + E_z = \frac{\hbar^2 \pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2).$$

